

STUDYHUB
CLASS 12 — MP BOARD
CHEMISTRY — SET 2
ORIGINAL SAMPLE QUESTION PAPER

Time: 3 Hours	Maximum Marks: 70
Medium: English	Set: 2

Note: This is an original StudyHub practice paper based on the MPBSE syllabus and examination pattern. It is not an official MPBSE paper.

GENERAL INSTRUCTIONS

1. All questions are compulsory.
2. Questions 1 to 5 are objective-type questions carrying a total of 28 marks.
3. Questions 6 to 12 carry 2 marks each.
4. Questions 13 to 16 carry 3 marks each.
5. Questions 17 to 20 carry 4 marks each.
6. Internal choices are provided from Question 6 to Question 20.
7. Write balanced chemical equations wherever necessary.
8. Use appropriate units in numerical problems.

SECTION - A : OBJECTIVE TYPE QUESTIONS

Q.1 Choose the correct option. (1×6=6)

(i) The packing efficiency of a face-centred cubic unit cell is approximately:

- A. 52.4%
- B. 68%
- C. 74%
- D. 78%

(ii) Which of the following is a colligative property?

- A. Viscosity
- B. Osmotic pressure
- C. Surface tension
- D. Refractive index

(iii) The unit of molar conductivity is:

- A. S cm^{-1}
- B. $\text{S cm}^2 \text{ mol}^{-1}$
- C. $\Omega \text{ cm}$
- D. $\Omega^{-1} \text{ cm}^{-1}$

(iv) For a zero-order reaction, the rate of reaction is:

- A. Dependent on concentration
- B. Independent of concentration
- C. Proportional to square of concentration
- D. Inversely proportional to concentration

(v) Which of the following ions has the highest number of unpaired electrons?

- A. Sc^{3+}
- B. Ti^{3+}
- C. Fe^{3+}
- D. Zn^{2+}

(vi) Which reagent is commonly used to distinguish aldehydes from ketones?

- A. Tollens' reagent
- B. Lucas reagent
- C. Grignard reagent

D. Conc. H_2SO_4

Q.2 Fill in the blanks. (1×6=6)

- (i) The defects in which equal numbers of cations and anions are missing from their normal lattice sites are called _____ defects.
- (ii) The pressure required to stop the flow of solvent molecules through a semipermeable membrane is called _____ pressure.
- (iii) The electrode potential of a cell under non-standard conditions is calculated using the _____ equation.
- (iv) The molecularity of an elementary reaction is always a _____ number.
- (v) The IUPAC name of CH_3COCH_3 is _____.
- (vi) The polymer obtained from tetrafluoroethene is called _____.

Q.3 Write True or False. (1×5=5)

- (i) Frenkel defect does not change the density of a crystal.
- (ii) Conductivity of an electrolytic solution generally decreases on dilution.
- (iii) The rate constant of a reaction depends on the concentration of reactants.
- (iv) Phenol is more acidic than ethanol.
- (v) Glucose is a reducing sugar.

Q.4 Match Column A with Column B. (1×6=6)

Column A	Column B
(i) Schottky defect	(a) CH_3COCH_3
(ii) Nernst equation	(b) Equal ionic vacancies
(iii) Acetone	(c) Cell potential
(iv) Tollen's reagent	(d) Silver mirror
(v) PVC	(e) Addition polymer
(vi) Amino acid	(f) Zwitterion

Q.5 Answer in one word/one sentence. (1×5=5)

- (i) What is the coordination number of an atom in a simple cubic unit cell?
- (ii) Name the instrument used to measure conductivity of an electrolytic solution.
- (iii) What is the minimum energy required to start a chemical reaction called?
- (iv) Name the reagent used for converting an alkyl halide into an alcohol by hydrolysis.
- (v) What is the monomer unit of natural rubber?

SECTION - B : VERY SHORT ANSWER QUESTIONS

Q.6 (2 Marks)

What is a unit cell? Write the relation between edge length and atomic radius for a simple cubic unit cell.

OR

Differentiate between crystalline and amorphous solids.

Q.7 (2 Marks)

What is osmotic pressure? Write one application of osmotic pressure.

OR

What is abnormal molar mass? Give one reason for its occurrence.

Q.8 (2 Marks)

What is meant by standard electrode potential?

OR

Why does the conductivity of an electrolytic solution change with dilution?

Q.9 (2 Marks)

Differentiate between order and molecularity of a reaction.

OR

Write the Arrhenius equation and explain the meaning of its symbols.

Q.10 (2 Marks)

Why do transition metals form alloys?

OR

Why do transition elements act as catalysts?

Q.11 (2 Marks)

Write the IUPAC name and structure of $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{OH}$.

OR

What happens when propanone reacts with hydrogen cyanide? Write the product formed.

Q.12 (2 Marks)

What is denaturation of proteins?

OR

What are biodegradable polymers? Give one example.

SECTION - C : SHORT ANSWER QUESTIONS

Q.13 (3 Marks)

Explain the three types of crystalline solids on the basis of intermolecular forces, giving one example of each.

OR

Calculate the mass of urea required to prepare 500 mL of a 0.2 M solution. Given: Molar mass of urea = 60 g mol^{-1} .

Q.14 (3 Marks)

Explain the working of an electrochemical cell and write the relation between Gibbs energy change and cell potential.

OR

Calculate the EMF of a cell if $E^\circ_{\text{cell}} = 1.10 \text{ V}$ and the reaction quotient $Q = 10$ at 298 K.

Q.15 (3 Marks)

Explain the factors affecting the rate of a chemical reaction.

OR

For a first-order reaction, the concentration of a reactant falls from 0.80 M to 0.20 M in 20 minutes. Calculate the rate constant.

Q.16 (3 Marks)

Explain the important properties of coordination compounds with respect to their colour, magnetic behaviour and stability.

OR

Write the IUPAC names of: (i) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (ii) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (iii) $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$.

SECTION - D : LONG ANSWER QUESTIONS

Q.17 (4 Marks)

Explain the preparation and properties of sodium carbonate by the Solvay process.

OR

Explain the preparation and properties of sodium hydroxide by the electrolysis of brine.

Q.18 (4 Marks)

Explain the important chemical properties of aldehydes and ketones. Write suitable equations for any four reactions.

OR

Explain Cannizzaro reaction with a suitable example. Why does it occur in aldehydes having no α -hydrogen?

Q.19 (4 Marks)

Explain the preparation of amines by the following methods: (i) Reduction of nitro compounds (ii) Gabriel phthalimide synthesis (iii) Hofmann bromamide reaction.

OR

Describe the basicity of amines and explain the factors affecting their basic character.

Q.20 (4 Marks)

Explain carbohydrates and classify them into monosaccharides, oligosaccharides and polysaccharides with suitable examples.

OR

Explain the classification of polymers on the basis of: (i) Source (ii) Structure (iii) Mode of polymerisation. Give suitable examples.

MARKS VERIFICATION

Questions	Marks
Q.1	6
Q.2	6
Q.3	5
Q.4	6
Q.5	5
Q.6-Q.12	14
Q.13-Q.16	12
Q.17-Q.20	16
TOTAL	70

6 + 6 + 5 + 6 + 5 + 14 + 12 + 16 = 70 Marks ✓

SHORT ANSWER KEY

Q.1: C, B, B, B, C, A. Q.2: Schottky; osmotic; Nernst; whole; propanone; Teflon. Q.3: True, False, False, True, True. Q.4: (i)-(b), (ii)-(c), (iii)-(a), (iv)-(d), (v)-(e), (vi)-(f). Q.5: 1; conductivity meter/conductometer; activation energy; aqueous KOH; isoprene.

Q.13: Mass of urea = 6 g. Q.14: Use the Nernst equation. Q.15: $k = (2.303/t) \log([R]_0/[R])$. Q.16: Use IUPAC coordination nomenclature. For Q.17-Q.20, award marks for correct concepts, equations, mechanisms, examples and explanations. Full solutions are not included.